Annex VI. External and Public Debt Sustainability

The rapid increase in domestic debt in recent years calls for strict adherence to the programmed fiscal consolidation path to preserve long-term debt sustainability. The September 2017 DSA confirms a moderate risk of debt distress for Benin.

A. Underlying Assumptions in the DSA

1. The assumptions in the baseline scenario are consistent with the medium-term macroeconomic framework underlying the program envisaged under the ECF arrangement. Key macroeconomic assumptions are as follows:

- **Global environment.** The nominal exchange rate (FCFA/USD) is assumed to appreciate slightly by about 3 percent over the baseline horizon and stabilize in the medium and long term. The external demand from Benin's trading partners is projected to be stable.
- **Growth impact.** Growth assumptions have remained roughly stable relative to the March 2017 DSA. Real GDP is expected to grow, on average, by 6.0 percent over the horizon 2017–19, supported by a a good performance of the agricultural sector, the ongoing recovery of the Nigerian economy, the scale up in public investment, the resilience of the tertiary sector (transport and telecommunications, trade), and an uptake in private investment in response to expected improvement in infrastructure. Growth is expected to stabilize at 6.6 percent in 2020-2022.
- **Inflation.** Capital goods will be partly imported, and the effect on non-tradable would be muted by high unemployment and labor mobility in WAEMU. Inflation is projected to average 2.8 percent on average in 2018–22, below the WAEMU convergence threshold of 3 percent.
- **Fiscal impact.** Tax revenue is projected to increase from 12.6 percent of GDP in 2016 to 14.8 percent of GDP in 2019 as the expected reforms in tax policy and administration mature. The primary deficit rises temporarily with higher capital spending and then turn into a surplus in 2021.
- **Current account impact.** As assumed in the March 2017 DSA, the current account deficit (including grants) is projected to peak in 2017, and would decline to 8.2 percent of GDP in 2018 and average 6.4 percent of GDP for 2019-22 as investment and import growth.
- **Financing.** The increase of central government investments of 9 percent of GDP is financed by concessional resources but also domestic financing. Non-concessional PPG debt financing is also included. Also, the recent rise of FDI in construction, manufacturing, and services are projected to continue, in line with Benin's recent achievements in improving its *Doing Business indicators* (ranking increased by 4 places from 155th ranking in 2017 to 151th ranking in 2018).

B. External Debt Sustainability Analysis

2. The results of the external DSA show that Benin's debt dynamics are sustainable under the baseline scenario, facing a moderate risk of debt distress (Tables A1, A2 and Figure A1). In the baseline, all debt indicators remain below their relevant policy-dependent thresholds. The present value (PV) of debt-to-GDP ratio, debt-to-exports ratio, debt-to-revenue ratio and debt service-to-exports ratio remain safely under the debt distress threshold under the baseline. The PV

of total PPG external debt is expected to rise from about 14.9 percent of GDP in 2017 to 15.8 percent of GDP on average for 2019–21, surging to 18.1 percent of GDP in 2037. The ratio would remain below the corresponding threshold of 40 percent of GDP throughout the projection period. Nonetheless, one indicator—the ratio of the PV of external debt to exports—exceeds its threshold in the case of an extreme shock to exports, while the debt-to-GDP ratio and all debt service indicators remain below thresholds. In the historical scenario, the ratios of the PV of debt-to-GDP plus remittances, PV of debt-to-exports plus remittances, and PV of debt-to-revenue show breaches to the thresholds. Thus, overall, Benin's risk of external debt distress is assessed to be moderate.

C. Public Debt Sustainability Analysis

3. Total public (external and domestic) debt is projected to rise during the scaling up of public investment and decline afterwards (Tables, A3 and Figure A2). The government's increased reliance on the regional financial market to finance public investment projects has increased the present value (PV) of public debt to 48.4 percent of GDP in 2017 (against 46.7 percent in the march 2017 DSA). Domestic public debt increased from about 8.6 percent of GDP to 33.6 percent of GDP between 2013 and 2017 to account for 60 percent of the total debt. Staff urged the authorities to remain steadfast in the implementation of the fiscal reforms to ensure that the programmed fiscal consolidation path is achieved to support the public debt anchor and preserve long-term debt sustainability. The PV of debt-to-GDP ratio is projected to rise from 41.9 percent in 2016 to 48.3 percent in 2017 with the surge in investment and then decline steadily. The ratio remains consistently below the indicative benchmark of 56 percent, a level that research has linked to increased probability of debt distress. The debt level also remains below the WAEMU convergence criteria of 70 percent of GDP. In the most extreme shock scenario (Figure A2), the peak PV of debt-to-GDP ratio exceeds 50 percent but remains below the 56 percent threshold. Overall, the dynamics in total public debt are consistent with a moderate risk of debt distress.

Table A1. Benin: External Debt Sustainability Framework, 2014–37 1/

(In percent of GDP, unless otherwise indicated)

	Actual			Historical Average 6/	Standard Deviation 6/	Projections									
	2014	2015	2016			2017	2018	2019	2020	2021	2022	2017-2022 Average	2027	2037	2023-2037 Average
External debt (nominal) 1/	**19.8**	**21.3**	**22.5**			**22.1**	**23.5**	**24.9**	**25.2**	**24.8**	**24.9**		**26.9**	**26.8**	
of which: public and publicly guaranteed (PPG)	19.8	21.3	22.5			22.1	23.5	24.9	25.2	24.8	24.9		26.9	26.8	
Change in external debt	3.1	1.5	1.2			-0.4	1.5	1.4	0.2	-0.4	0.2		0.4	-0.4	
Identified net debt-creating flows	3.6	10.2	5.3			3.4	1.4	-1.9	-2.2	-2.5	-2.9		0.4	-0.4	
Non-interest current account deficit	**8.4**	**7.8**	**7.1**	**7.6**	**0.5**	**8.8**	**8.0**	**6.7**	**6.2**	**6.0**	**5.5**		**3.8**	**2.4**	3.7
Deficit in balance of goods and services	15.2	12.2	11.7			13.6	12.6	11.7	11.5	10.9	10.3		7.9	5.6	
Exports	15.8	18.0	15.9			17.3	18.3	19.0	18.0	18.8	18.7		23.4	28.0	
Imports	31.0	30.1	27.6			30.9	30.9	30.7	29.5	29.7	28.9		31.4	33.6	
Net current transfers (negative = inflow)	-6.4	-4.0	-4.3	-4.7	0.9	-4.6	-4.5	-4.9	-5.1	-4.8	-4.7		-3.7	-2.8	-3.5
of which: official	-4.5	-1.6	-1.7			-1.8	-1.6	-1.9	-1.7	-1.5	-1.5		-1.5	-1.5	
Other current account flows (negative = net inflow)	-0.4	-0.3	-0.3			-0.2	-0.1	-0.1	-0.1	-0.1	-0.1		-0.4	-0.5	
Net FDI (negative = inflow)	**-4.0**	**-1.4**	**-1.4**	**-2.5**	**1.1**	**-4.6**	**-5.8**	**-7.6**	**-7.3**	**-7.3**	**-7.3**		**-2.6**	**-2.0**	-2.2
Endogenous debt dynamics 2/	**-0.8**	**3.8**	**-0.5**			**-0.8**	**-0.8**	**-1.0**	**-1.1**	**-1.3**	**-1.0**		**-0.8**	**-0.8**	
Contribution from nominal interest rate	0.2	0.4	0.2			0.4	0.4	0.4	0.4	0.4	0.4		0.4	0.5	
Contribution from real GDP growth	-1.0	-0.5	-0.8			-1.2	-1.2	-1.4	-1.5	-1.6	-1.4		-1.2	-1.2	
Contribution from price and exchange rate changes	0.0	3.9	0.1			...	...	...	...	...	...		...	...	
Residual (3-4) 3/	**-0.5**	**-8.7**	**-4.1**			**-3.8**	**0.1**	**3.3**	**2.4**	**2.2**	**3.0**		**0.0**	**0.0**	
of which: exceptional financing	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
PV of external debt 4/	...	...	14.9			14.9	15.4	15.8	15.9	15.7	15.8		17.2	18.1	
In percent of exports	...	...	94.0			85.7	84.4	83.5	88.2	83.4	84.7		73.4	64.7	
PV of PPG external debt	**...**	**...**	**14.9**			**14.9**	**15.4**	**15.8**	**15.9**	**15.7**	**15.8**		**17.2**	**18.1**	
In percent of exports	**...**	**...**	**94.0**			**85.7**	**84.4**	**83.5**	**88.2**	**83.4**	**84.7**		**73.4**	**64.7**	
In percent of government revenues	**...**	**...**	**101.7**			**88.3**	**88.3**	**91.4**	**88.8**	**85.1**	**84.0**		**89.4**	**91.8**	
Debt service-to-exports ratio (in percent)	**5.1**	**5.3**	**4.7**			**5.5**	**6.0**	**5.8**	**6.1**	**5.5**	**5.2**		**4.1**	**4.5**	
PPG debt service-to-exports ratio (in percent)	**5.1**	**5.3**	**4.7**			**5.5**	**6.0**	**5.8**	**6.1**	**5.5**	**5.2**		**4.1**	**4.5**	
PPG debt service-to-revenue ratio (in percent)	**5.0**	**5.7**	**5.1**			**5.6**	**6.3**	**6.3**	**6.1**	**5.6**	**5.1**		**5.0**	**6.4**	
Total gross financing need (Billions of U.S. dollars)	0.5	0.6	0.6			0.5	0.3	0.0	0.0	0.0	-0.1		0.5	0.7	
Non-interest current account deficit that stabilizes debt ratio	5.3	6.4	5.9			9.2	6.5	5.3	6.0	6.4	5.3		3.4	2.8	
Key macroeconomic assumptions															
Real GDP growth (in percent)	6.4	2.1	4.0	4.3	1.9	5.6	6.0	6.3	6.7	7.1	6.2	6.3	4.8	4.8	5.0
GDP deflator in US dollar terms (change in percent)	-0.2	-16.4	-0.6	1.3	8.7	2.3	6.6	3.1	2.7	2.8	3.4	3.5	2.0	2.0	2.1
Effective interest rate (percent) 5/	1.3	1.5	1.2	3.0	4.6	1.8	2.1	1.8	1.7	1.6	1.6	1.8	1.6	1.8	1.7
Growth of exports of G&S (US dollar terms, in percent)	8.4	-3.3	-8.7	9.9	21.5	18.2	19.3	13.6	4.3	14.7	8.9	13.2	7.0	8.0	10.2
Growth of imports of G&S (US dollar terms, in percent)	16.5	-17.2	-5.2	9.5	21.7	21.0	13.0	8.8	5.4	10.8	7.0	11.0	7.5	7.5	8.3
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	44.9	44.6	44.4	44.8	45.2	45.4	44.9	39.5	34.9	38.0
Government revenues (excluding grants, in percent of GDP)	16.3	16.7	14.7			16.8	17.5	17.3	17.9	18.4	18.8		19.2	19.7	19.4
Aid flows (in Billions of US dollars) 7/	0.1	0.0	0.1			0.1	0.1	0.2	0.2	0.1	0.2		0.1	0.1	
of which: Grants	0.1	0.0	0.1			0.1	0.1	0.2	0.2	0.1	0.2		0.1	0.1	
of which: Concessional loans	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Grant-equivalent financing (in percent of GDP) 8/	...	...	...			3.4	3.1	3.3	2.9	2.2	2.4		1.5	1.0	1.4
Grant-equivalent financing (in percent of external financing) 8/	...	...	...			57.9	59.5	62.1	63.0	60.6	59.0		48.9	42.9	47.1
Memorandum items:															
Nominal GDP (Billions of US dollars)	9.7	8.3	8.6			9.3	10.5	11.5	12.6	13.9	15.2		22.2	43.2	
Nominal dollar GDP growth	6.1	-14.7	3.4			8.1	13.1	9.6	9.6	10.0	9.8	10.0	6.9	6.9	7.2
PV of PPG external debt (in Billions of US dollars)			1.2			1.4	1.6	1.8	2.0	2.2	2.4		3.8	7.8	
(PVt-PVt-1)/GDPt-1 (in percent)						2.5	2.0	1.9	1.6	1.4	1.7	1.8	1.5	1.1	1.4
Gross workers' remittances (Billions of US dollars)	0.3	0.3	0.3			0.3	0.4	0.4	0.5	0.5	0.5		0.5	0.7	
PV of PPG external debt (in percent of GDP + remittances)	...	...	14.4			14.3	14.9	15.2	15.2	15.2	15.3		16.8	17.9	
PV of PPG external debt (in percent of exports + remittances)	...	...	77.2			71.0	70.4	69.5	71.1	70.4	71.4		66.4	61.4	
Debt service of PPG external debt (in percent of exports + remittances)	...	...	3.9			4.5	5.0	4.8	4.9	4.7	4.4		3.7	4.3	

Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Assumes that PV of private sector debt is equivalent to its face value.

5/ Current-year interest payments divided by previous period debt stock.

6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

7/ Defined as grants, concessional loans, and debt relief.

8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table A2. Benin: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2017–37

(In percent)

	Projections							
	2017	2018	2019	2020	2021	2022	2027	2037
PV of debt-to GDP ratio								
Baseline	15	15	16	16	16	16	17	18
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2017-2037 1/	15	18	23	28	32	37	48	61
A2. New public sector loans on less favorable terms in 2017-2037 2	15	16	18	19	19	20	24	30
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2018-2019	15	16	17	17	17	17	19	20
B2. Export value growth at historical average minus one standard deviation in 2018-2019 3/	15	19	24	24	23	23	23	20
B3. US dollar GDP deflator at historical average minus one standard deviation in 2018-2019	15	18	20	20	20	20	22	23
B4. Net non-debt creating flows at historical average minus one standard deviation in 2018-2019 4/	15	19	24	24	23	23	23	20
B5. Combination of B1-B4 using one-half standard deviation shocks	15	23	36	35	34	34	33	27
B6. One-time 30 percent nominal depreciation relative to the baseline in 2018 5/	15	22	22	22	22	22	24	25
PV of debt-to-exports ratio								
Baseline	86	84	84	88	83	85	73	65
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2017-2037 1/	86	100	122	153	170	195	204	215
A2. New public sector loans on less favorable terms in 2017-2037 2	86	90	94	103	101	106	103	105
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2018-2019	86	85	84	88	84	85	73	65
B2. Export value growth at historical average minus one standard deviation in 2018-2019 3/	86	137	218	226	211	210	169	125
B3. US dollar GDP deflator at historical average minus one standard deviation in 2018-2019	86	85	84	88	84	85	73	65
B4. Net non-debt creating flows at historical average minus one standard deviation in 2018-2019 4/	86	103	126	131	122	121	97	72
B5. Combination of B1-B4 using one-half standard deviation shocks	86	136	211	217	201	199	155	106
B6. One-time 30 percent nominal depreciation relative to the baseline in 2018 5/	86	85	84	88	84	85	73	65
PV of debt-to-revenue ratio								
Baseline	88	88	91	89	85	84	89	92
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2017-2037 1/	88	105	133	154	173	195	248	305
A2. New public sector loans on less favorable terms in 2017-2037 2	88	94	102	104	103	106	126	149
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2018-2019	88	92	98	95	92	91	96	99
B2. Export value growth at historical average minus one standard deviation in 2018-2019 3/	88	106	137	131	124	121	118	102
B3. US dollar GDP deflator at historical average minus one standard deviation in 2018-2019	88	102	117	114	109	108	115	118
B4. Net non-debt creating flows at historical average minus one standard deviation in 2018-2019 4/	88	108	138	131	124	121	118	102
B5. Combination of B1-B4 using one-half standard deviation shocks	88	133	206	195	184	178	169	134
B6. One-time 30 percent nominal depreciation relative to the baseline in 2018 5/	88	123	127	124	119	118	124	128

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.

3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.

Table A3. Benin: Public Sector Debt Sustainability Framework, 2014–37

(In percent of GDP, unless otherwise indicated)

	Actual					Estimate	Projections								
	2014	2015	2016	Average 5/	Standard Deviation 5/	2017	2018	2019	2020	2021	2022	2017-22 Average	2027	2037	2023-37 Average
Public sector debt 1/	30.5	42.4	49.5			55.6	56.2	53.5	48.9	44.6	40.6		33.6	33.5	
of which: foreign-currency denominated	19.8	21.3	22.5			22.1	23.3	24.6	24.9	24.5	24.9		27.1	27.0	
Change in public sector debt	5.2	12.0	7.1			6.1	0.6	-2.8	-4.6	-4.3	-4.0		0.4	-0.4	
Identified debt-creating flows	2.2	10.0	5.2			1.1	0.1	-3.0	-4.1	-4.4	-4.5		-1.4	-0.7	
Primary deficit	1.5	7.3	4.8	2.1	2.5	4.1	2.2	-0.6	-1.5	-1.9	-2.5	0.0	-0.4	0.3	-0.3
Revenue and grants	17.2	17.3	15.3			18.2	19.1	19.2	19.6	19.5	19.9		19.9	20.2	
of which: grants	0.9	0.6	0.7			1.4	1.6	1.9	1.6	1.0	1.0		0.5	0.3	
Primary (noninterest) expenditure	18.7	24.6	20.1			22.3	21.3	18.6	18.1	17.6	17.4		19.5	20.5	
Automatic debt dynamics	0.7	2.7	0.4			-3.0	-2.1	-2.4	-2.5	-2.6	-2.0		-1.1	-1.1	
Contribution from interest rate/growth differential	-1.1	0.1	-0.3			-0.8	-2.0	-2.3	-2.5	-2.6	-2.0		-1.1	-1.1	
of which: contribution from average real interest rate	0.4	0.7	1.4			1.9	1.1	1.0	0.8	0.7	0.6		0.4	0.5	
of which: contribution from real GDP growth	-1.5	-0.6	-1.6			-2.6	-3.2	-3.4	-3.4	-3.2	-2.6		-1.5	-1.6	
Contribution from real exchange rate depreciation	1.8	2.6	0.7			-2.2	-0.1	-0.1	0.0	0.0	0.0		...	...	
Other identified debt-creating flows	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	2.9	2.0	1.9			5.0	0.5	0.2	-0.5	0.1	0.5		1.8	0.3	
Other Sustainability Indicators															
PV of public sector debt	...	...	41.9			48.4	48.3	44.7	39.9	35.8	31.7		23.8	24.8	
of which: foreign-currency denominated	...	...	14.9			14.9	15.4	15.8	15.9	15.7	15.9		17.3	18.3	
of which: external	...	...	14.9			14.9	15.4	15.8	15.9	15.7	15.9		17.3	18.3	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	8.1	14.7	12.1			13.8	12.5	9.6	7.5	5.9	4.1		2.9	3.7	
PV of public sector debt-to-revenue and grants ratio (in percent)	...	...	273.2			265.4	253.2	232.4	203.8	183.9	159.7		119.9	122.7	
PV of public sector debt-to-revenue ratio (in percent)	...	...	285.9			287.6	276.4	257.5	222.5	194.0	168.1		122.9	124.5	
of which: external 3/	...	...	101.7			88.3	88.3	91.2	88.6	85.0	84.5		89.4	91.9	
Debt service-to-revenue and grants ratio (in percent) 4/	11.7	14.4	18.2			36.9	42.8	42.5	37.1	32.0	27.0		9.3	9.6	
Debt service-to-revenue ratio (in percent) 4/	12.3	14.9	19.0			40.0	46.7	47.0	40.5	33.8	28.4		9.5	9.8	
Primary deficit that stabilizes the debt-to-GDP ratio	-3.6	-4.7	-2.3			-2.0	1.6	2.2	3.0	2.4	1.5		-0.7	0.8	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	6.4	2.1	4.0	4.3	1.9	5.6	6.0	6.3	6.7	7.1	6.2	6.3	4.8	4.8	5.0
Average nominal interest rate on forex debt (in percent)	1.3	1.5	1.2	3.0	4.6	1.8	2.1	1.8	1.7	1.6	1.6	1.8	1.6	1.8	1.7
Average real interest rate on domestic debt (in percent)	2.3	3.9	5.3	0.9	3.7	6.2	3.6	3.9	3.8	4.2	4.4	4.4	8.8	8.8	8.1
Real exchange rate depreciation (in percent, + indicates depreciation)	11.1	13.4	3.2	2.6	7.9	-10.4	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	-0.2	0.1	-0.3	2.3	2.8	0.4	2.3	2.6	2.7	2.8	2.6	2.2	2.0	2.0	2.1
Growth of real primary spending (deflated by GDP deflator, in percent)	-3.4	34.2	-14.9	1.6	12.4	17.2	1.1	-7.0	3.6	4.2	4.8	4.0	10.0	4.8	6.2
Grant element of new external borrowing (in percent)	...	...	...	...	...	44.9	44.6	44.4	44.8	45.2	45.3	44.9	39.5	34.9	...

Sources: Country authorities; and staff estimates and projections.

1/ [Indicate coverage of public sector, e.g., general government or nonfinancial public sector. Also whether net or gross debt is used.]

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table A4. Benin: Sensitivity Analysis for Key Indicators of Public Debt 2017–37

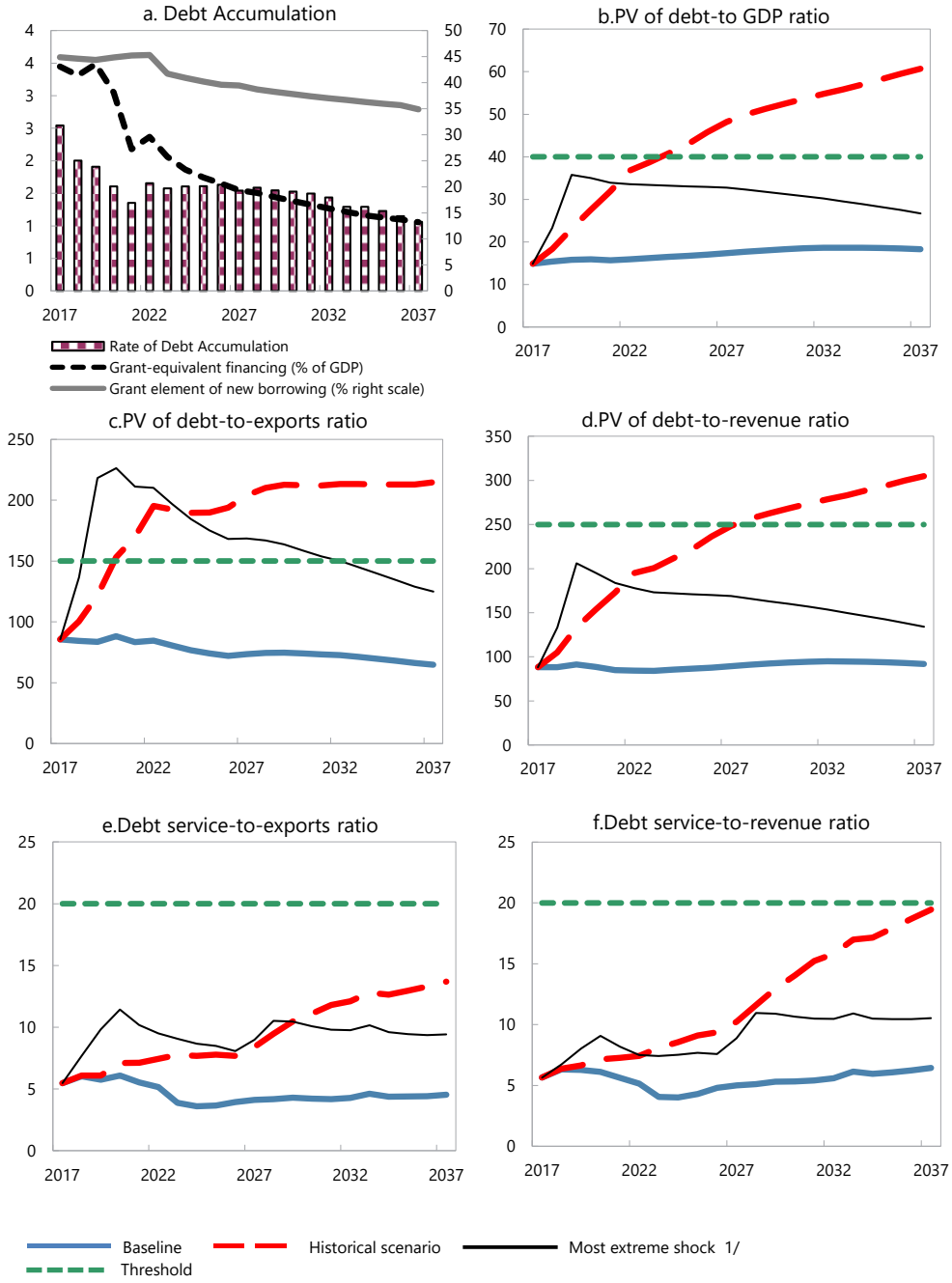
	Projections							
	2017	2018	2019	2020	2021	2022	2027	2037
PV of Debt-to-GDP Ratio								
Baseline	48	48	45	40	36	32	24	25
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	48	49	48	46	45	44	48	55
A2. Primary balance is unchanged from 2017	48	49	49	47	46	46	52	65
A3. Permanently lower GDP growth 1/	48	49	45	41	37	33	27	35
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2018-2019	48	50	49	45	41	38	33	40
B2. Primary balance is at historical average minus one standard deviations in 2018-2019	48	50	49	44	40	36	27	27
B3. Combination of B1-B2 using one half standard deviation shocks	48	50	50	46	42	38	32	37
B4. One-time 30 percent real depreciation in 2018	48	54	49	43	39	34	24	24
B5. 10 percent of GDP increase in other debt-creating flows in 2018	48	55	51	46	41	37	28	27
PV of Debt-to-Revenue Ratio 2/								
Baseline	265	253	232	204	184	160	120	123
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	265	257	248	235	232	222	241	269
A2. Primary balance is unchanged from 2017	265	259	254	241	239	232	264	322
A3. Permanently lower GDP growth 1/	265	254	235	207	189	166	136	175
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2018-2019	265	263	254	228	211	188	164	197
B2. Primary balance is at historical average minus one standard deviations in 2018-2019	265	261	257	226	205	180	137	133
B3. Combination of B1-B2 using one half standard deviation shocks	265	263	260	233	214	191	161	183
B4. One-time 30 percent real depreciation in 2018	265	282	255	222	199	172	122	120
B5. 10 percent of GDP increase in other debt-creating flows in 2018	265	286	263	232	211	185	141	136
Debt Service-to-Revenue Ratio 2/								
Baseline	37	43	42	37	32	27	9	10
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	37	43	44	39	35	31	13	19
A2. Primary balance is unchanged from 2017	37	43	43	38	33	29	13	20
A3. Permanently lower GDP growth 1/	37	43	43	38	33	28	10	12
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2018-2019	37	44	45	40	35	29	11	13
B2. Primary balance is at historical average minus one standard deviations in 2018-2019	37	43	43	38	33	28	10	10
B3. Combination of B1-B2 using one half standard deviation shocks	37	44	45	40	34	29	11	13
B4. One-time 30 percent real depreciation in 2018	37	44	45	40	34	29	12	13
B5. 10 percent of GDP increase in other debt-creating flows in 2018	37	43	43	38	33	28	10	11

Sources: Country authorities; and staff estimates and projections.

1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.

2/ Revenues are defined inclusive of grants.

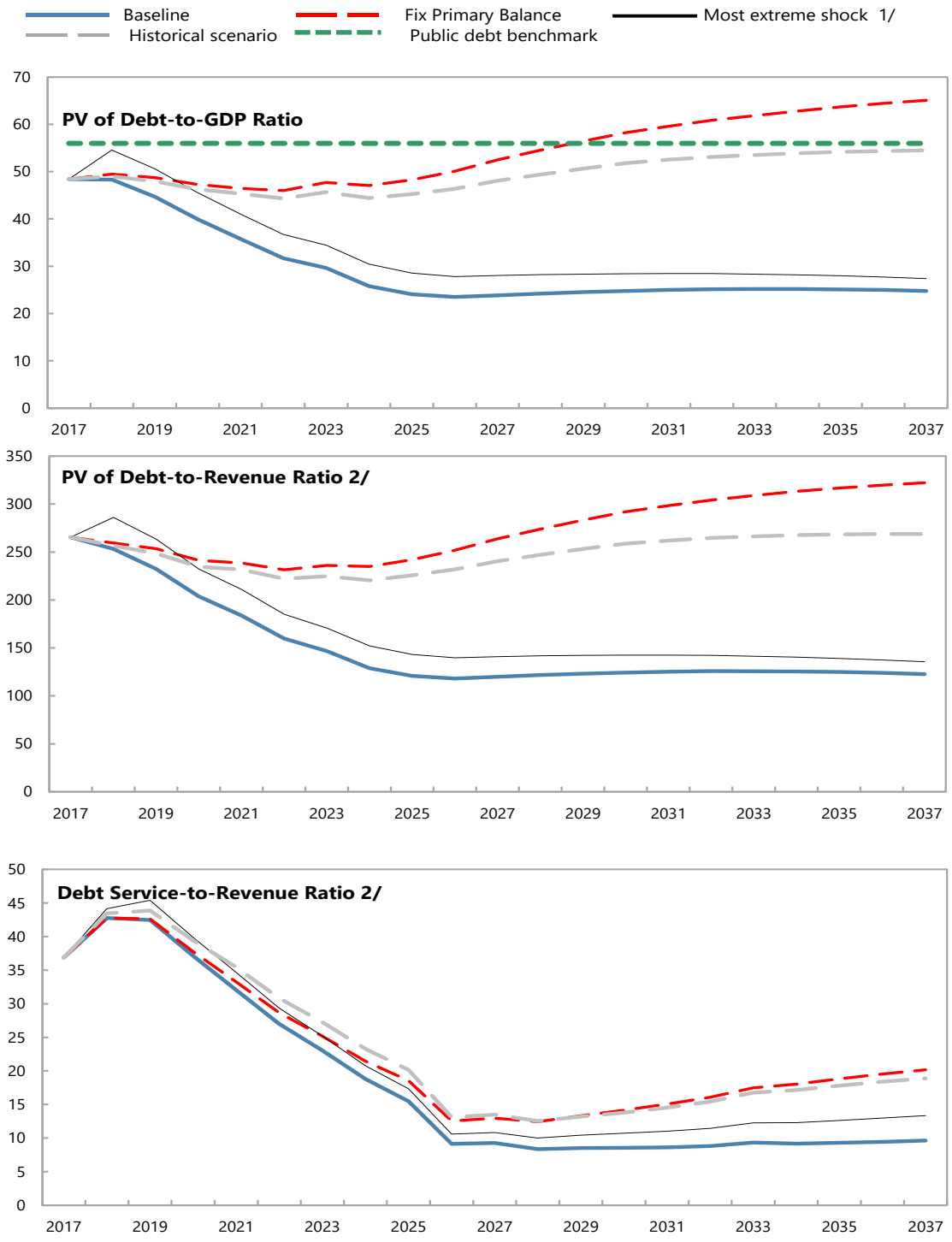
Figure A1. Benin: Indicators of Public and Publicly Guaranteed External Debt, 2017–37 1/



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio on or before 2027. In figure b. it corresponds to a Combination shock; in c. to a Exports shock; in d. to a Combination shock; in e. to a Exports shock and in figure f. to a Combination shock

Figure A2. Benin: Indicators of Public Debt Under Alternative Scenarios, 2017–37 1/



Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio on or before 2027.
2/ Revenues are defined inclusive of grants.